

Breast Cancer Prediction Model

Department of Computer Science & Information Technology
University of South Asia, Lahore

Hafiz Abdul Sami

Department of Computer Science & Information
Technology, University of South Asia, Lahore

Sohaib Ramzan

Department of Computer Science & Information
Technology, University of South Asia, Lahore

Kanwar. Muhammad Shaheen

Department of Computer Science & Information
Technology, University of South Asia, Lahore

Zain Ul Abadeen

Department of Computer Science and Information
Technology, University of South Asia, Lahore

Abstract

Breast cancer is a prevalent disease that has a potential influence on the lives of countless women globally. Early diagnosis and intervention are crucial for successful treatment and better patient outcomes. Machine learning algorithms have shown promising results in developing accurate and dependable prediction models for breast cancer. In this research, we conduct an extensive overview of various machine learning (ML) techniques employed to develop breast cancer prediction models using diverse datasets. Our study explores the literature on several ML algorithms utilized for breast cancer prediction.

Breast cancer is one of the most common cancers among women worldwide. Early detection and diagnosis are crucial for effective treatment and improving survival rates. The Breast Cancer Wisconsin (Diagnostic) dataset contains features computed from digitized images of breast cancer biopsies. This dataset is used to train a machine learning model to predict the malignancy of tumors based on these features.

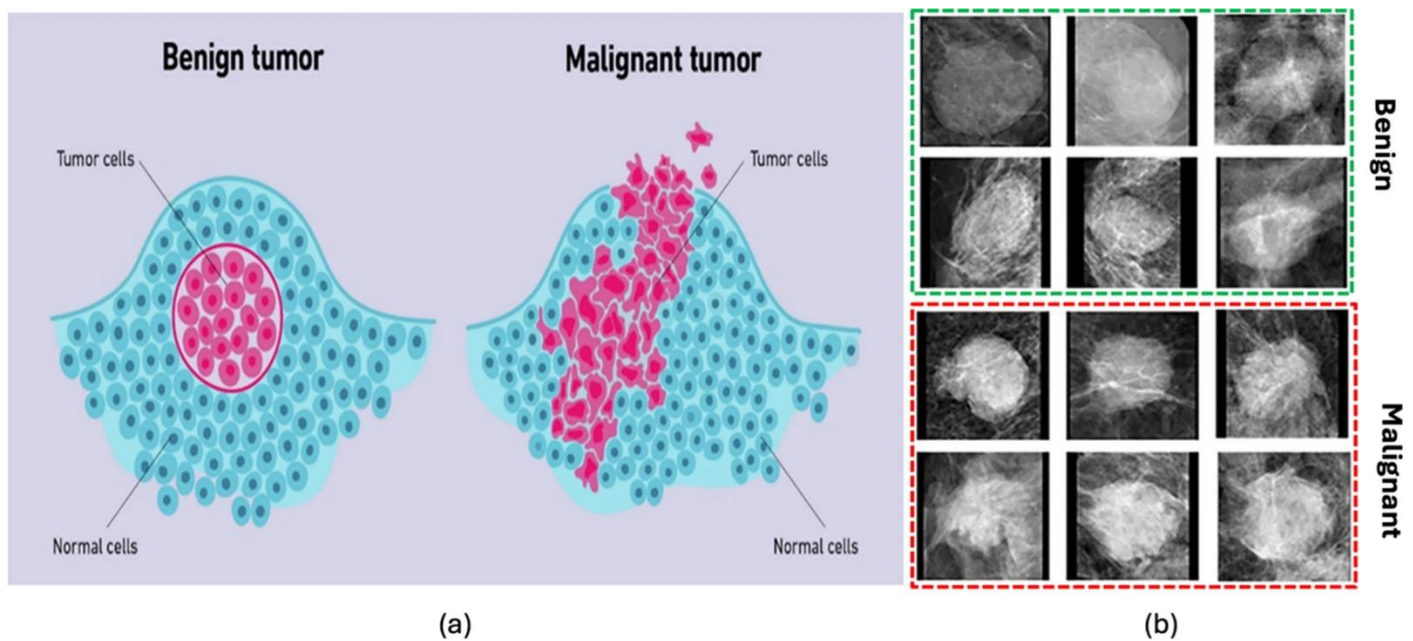
This **Breast Cancer Prediction Model** is a machine learning project aimed at predicting whether a breast tumor is malignant or benign based on various diagnostic features. This project leverages data science and machine learning techniques to build a predictive model and deploy it as an interactive web application using Streamlit. Building a model from scratch, loading the data, data analysis, data cleaning and preparing the data for the algorithm, data visualization, building models, prediction with the help of multiple ML Classification algorithms like KNN, SVM, Random Forest, Decision Tree, Logistic Regression.

With *Logistic Regression Test Set Accuracy: 99.12%, Random Forest Test Set Accuracy: 96.49%, SVM Test Set Accuracy: 97.37%, KNN Test Set Accuracy: 94.74%, Decision Tree Test Set Accuracy: 92.98%.*

Keywords: *Breast Cancer Prediction, Diagnostic features, Machine Learning, Data Science, Data Analytics, Benign, Malignant, Early Diagnosis of Cancer.*

I. Introduction

Breast cancer begins when some cells in the breast start to grow uncontrollably, forming a mass called a tumor. A breast cancer diagnosis typically falls into one of two main categories—benign (non-cancerous) or malignant (cancerous). Malignant tumors are dangerous as they can spread to distant sites in the body through the bloodstream or lymph system, a process known as metastasis^{2,3}. Figure 1(a) illustrates the distinction between benign and malignant tumors in terms of the normal cells and tumor cells, and (b) shows the benign and malignant masses. Benign tumors generally stay localized in one area and do not metastasize. Breast cancer manifests through several symptoms—a noticeable lump or mass in the breast, changes in breast size or shape compared to the other breast, alterations in the skin overlying the breast like dimpling or puckering, newly inverted nipple, redness or scaliness of breast skin, breast pain, and nipple discharge other than breast milk.



Visualization of breast cancer: (a) benign and malignant tumor cells, (b) benign and malignant masses.

Breast cancer is the second largest killer of women after cardiovascular disease. And more than 8% of women will experience it. Every year, more than 500,000 women are diagnosed with breast cancer, as stated in the World Health Organization's annual report. In developing countries, due to a lack of screening programs and awareness, women often present at an advanced stage where treatment options are limited. Known risk factors for breast cancer include genetic mutations in BRCA genes, reproductive history (nulliparity, early menarche, late menopause), hormonal factors (use of hormone replacement therapy, oral contraceptives), obesity, alcohol consumption, smoking, radiation exposure at a young age and family history.

Many people were affected by cancer during this period. We can't pinpoint the origin of the sickness since it's tied to factors beyond our control. This is also a screening technique for identifying the cancer's aggressiveness. Several assessment items are connected to cancer detection, including clump thickness, cell size consistency, and shape regularity. The outcome is difficult even for those tasked with inspiring others to take action, and yet the use of machine learning and other computer science techniques as general diagnostic tools has expanded in recent years. Countless numbers of people's lives have been saved by computer diagnostic programs that use diseases that have killed millions. In the realm of surgery, robotics is indispensable. Aside from other artificial intelligence's widespread usage in cancer detection, the system deployed in the intensive care unit is highly effective. One in eight American females may get cancer between the ages of 15 and 19. Breast cancer is the result of unchecked cell division, which can also cause breasts to sag (called tumors). In most cases, the tumor poses no health risk. The necessity for precise categorization in the clinic may be a severe challenge for doctors and health workers, especially when the correct identification of the determinants might contribute to survival, regardless of whether the condition is benign or malignant.

The diagnosis of breast cancer involves a step-wise approach starting with a thorough clinical examination and radiological tests like mammograms and breast ultrasounds. This may be followed by tissue sampling through fine needle aspiration cytology (FNAC) or biopsy from suspicious areas and microscopic assessment to confirm malignancy. As symptoms of breast cancer can be nonspecific with wide variation across patients, the combination of these investigations is needed for accurate diagnosis in the majority of cases. So, to monitor and diagnose diseases, a human observer must be able to pick out very particular signal features. Due to the large number of patients in the critical care unit and the need for round-the-clock monitoring, several CAD approaches for computer-aided medical systems have emerged in the recent decade to meet this issue. With these strategies, the challenge of classifying quantitative features may be posed rather than relying on qualitative diagnostic criteria. Machine learning algorithms can predict breast cancer diagnosis and prognosis. The purpose of this work is to evaluate the efficacy and performance of these algorithms in terms of their accuracy, sensitivity, range, and precision. In the past 25 years, the importance of artificial intelligence has increased. As scientists realize the importance of making firm decisions about how to treat certain diseases, the use of computers and machine learning as diagnostic tools has become deadly, which is the most serious disease screening task in the medical field. One of the most important functions of disease is the definition of cancer. Using machine learning technology, doctors can detect, identify, and classify tumors as benign or malignant. There are some challenges in analyzing patient data and choosing doctors and specialists, but cognitive systems and computational methods (such as ML for classification) will ultimately help doctors and professionals. However, as machine learning models become more complex, there is a need for Explainable AI (XAI) techniques to interpret these models and understand how they arrive at their predictions.

In this context, the application of machine learning (ML) methods has emerged as a promising approach for analyzing large-scale cancer patient data to identify hidden patterns, uncover significant biomarkers, and improve predictive accuracy. ML algorithms such as decision trees, support vector machines (SVMs), and neural networks are particularly well-suited to analyzing high-dimensional datasets, such as those obtained from electronic health records (EHRs), gene expression profiles, and medical imaging, where traditional statistical approaches may falter.

Recent studies have demonstrated that machine learning can significantly enhance the prediction of cancer-related outcomes. For example, a study utilizing random forest classifiers on breast cancer data achieved a classification accuracy of 95.8%, demonstrating the model's ability to differentiate between benign and malignant cases.

The potential of ML to revolutionize breast cancer prediction models cannot be overstated, as these advanced algorithms offer a level of precision and personalization previously unattainable.

At the end, breast cancer remains a pressing global health issue, with its prevalence and impact on society continuing to rise. It stands as the most frequently diagnosed cancer among women, presenting a complex challenge to healthcare systems worldwide. The increasing incidence rates, as reported by the World Health Organization, highlight an urgent need for more efficient and accurate detection and treatment methods. The alarming statistics underscore the necessity for continual research and development in this area, with a focus on advanced solutions that can adapt to the evolving nature of this disease.

II. Literature Review

In the literature, many works applied Artificial Intelligence (AI) techniques for breast cancer diagnosis to improve classification accuracy and time response. In this section, we give some works related to the solution of medical breast cancer diagnosis using machine learning and deep learning approaches.

Arpit B. and Aruna T. proposed a genetically optimized neural network (GONN) for breast cancer classification (malignant and benign). They optimized the neural network architecture by introducing new crossover and mutation operators. To evaluate their work, they used WBCD and compared the classification accuracy, sensitivity, specificity, confusion matrix, ROC curves, and AUC under ROC curves of GONN with the classical model. This method presents a good accuracy classification. However, it can be improved by using a larger dataset than WBCD, feature extraction to make GONN more efficient for real-time diagnosis of Breast Cancer.

Ashraf O. I. and Siti M. S. proposed a computer-based method to classify automatically breast cancer disease. The method applied a multilayer perceptron (MLP) neural network based on an enhanced non-dominated sorting genetic algorithm (NSGA-II) to optimize both the accuracy and network structure. Compared to other methods, this work improves classification accuracy. However, MLP can get stuck in local minima.

Na L. et al. proposed an intelligent classification model for breast cancer diagnosis based on a hybrid feature selection approach: gain-directed simulated annealing genetic algorithm wrapper (IGSAGAW), to remove redundant and irrelevant features from the feature space and cost-sensitive support vector machine (CSSVM) learning algorithm. This process can improve the classification accuracy and reduce the computational cost. The proposed method is applied to Wisconsin Original Breast Cancer (WBC) and WBCD to verify its effectiveness. The proposed work shows a good performance and decreases the calculation complexity.

Nawel Z. et al. presented a conception and implementation of Computer-Assisted Detection (CAD) for mammogram image classification. The system is based on a GA-based features selection algorithm to reduce the dimensionality of the feature vector and a semi-supervised support vector machine (S3VM) for classification. Experiments were validated on the Digital Database for Screening Mammography (DDSM) dataset. The proposed approach improved accuracy. **Abdulkader H. et al.** developed an automated system for the classification of breast tissue. The system uses two machine learning techniques: Feed forward neural network using the back propagation learning algorithm (BPNN) and radial basis function network (RBFN). Breast cancer tissues were classified into 6 different tissues: Carcinoma, Fibro-adenoma, Mastopathy, Glandular, Connective, and Adipose tissue. Data were acquired using an electrical impedance spectroscopy (EIS) method. The Radial basis function network outperformed the back propagation network for classifying six different breast tissues in terms of accuracy, minimum error, maximum epochs, and training time. The proposed system improved accuracy and decreased training time. However, learning with neural networks can be weak in generalizing and can get stuck in local optima. **Haifeng W. et al.** designed an SVM-based ensemble learning model for breast cancer diagnosis. The proposed ensemble model includes two types of SVM structures, i.e., a C-SVM and an SVM, and six types of kernel functions. To import the expertise of different base classifiers on diagnostic tasks, a Weighted Area Under the Receiver Operating Characteristic Curve Ensemble (WAUCE) mechanism is proposed for model hybridization. The model was evaluated based on two datasets: the Wisconsin Breast Cancer (WBC) dataset and the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, and one large dataset, the Surveillance, Epidemiology, and End Results (SEER) dataset. The proposed model increases diagnosis accuracy compared to other works based on a single SVM. However, it is a computationally expensive method, and the training time is high.

Kemal P. et al. proposed a hybrid approach based on MAD normalization, KMC-based feature weighting, and an AdaBoostM1 classifier. The detection of the presence of breast cancer is done in three steps: In the first step, the dataset was first normalized by the MAD normalization method. In the second step, k-means clustering (KMC) based feature weighting has been used for weighting the normalized data. Finally, the AdaBoostM1 classifier has been used to classify the weighted dataset. The Breast Cancer Coimbra dataset (BCC), taken from the UCI machine learning database, was used. This method shows good results in terms of accuracy. However, it is a computationally expensive method.

Teresa A. J. et al. proposed a classification method of hematoxylin and eosin-stained breast biopsy images using Convolutional Neural Networks (CNNs). They provide four classes of medical relevance: normal tissue, benign lesion, in situ carcinoma, and invasive carcinoma. The proposed CNN architecture is designed to integrate information from multiple histological scales. The model is applied to an image dataset composed of high-resolution, uncompressed, and annotated HE stain images from the Bioimaging 2015 breast histology classification challenge.

Fabio A. et al. used a deep learning approach to classify breast cancer histopathological images from BreaKHis, a public dataset. They proposed a method based on the extraction of image patches for training the Convolutional Neural Network (CNN) and the combination of these patches for final classification. This method allows for avoiding adaptations of the model that can lead to a more complex and computationally costly architecture. However, the lack of data results in costly and expensive experimentations.

Hiba A. et al. presented a comparison between the performances of four classifiers: Support Vector Machine (SVM), Decision Tree (C4.5), Naive Bayes (NB), and k Nearest Neighbors (k-NN) on the *Wisconsin Breast Cancer datasets*.

Summary of Contributions in Breast Cancer Diagnosis Using AI

Author(s)	Method/Model	Dataset(s)	Key Contribution	Limitations
Arpit B. & Aruna T.	Genetically Optimized Neural Network (GONN)	WBCD	Used new crossover and mutation operators for NN optimization. Good accuracy, ROC, and AUC.	Needs a larger dataset and feature extraction for real-time efficiency.
Ashraf O. I. & Siti M. S.	MLP Neural Network optimized with NSGA-II	Not specified	Enhanced network structure and accuracy using NSGA-II.	MLP can get stuck in local minima.
Na L. et al.	Hybrid Feature Selection (IGSAGAW) + Cost-Sensitive SVM (CSSVM)	WBC, WBCD	Improved classification accuracy and reduced computational cost.	Not explicitly mentioned.
Nawel Z. et al.	GA-based Feature Selection + Semi-supervised SVM (S3VM)	DDSM	CAD system for mammogram classification with improved accuracy.	Not specified.
Abdulkader H. et al.	BPNN and RBFN using Electrical Impedance Spectroscopy (EIS)	EIS-based data	Classified 6 tissue types; RBFN outperformed BPNN in accuracy and training time.	NN models may generalize poorly and get stuck in local optima.
Haifeng W. et al.	SVM-based Ensemble (C-SVM, SVM, 6 Kernels) + WAUCE	WBC, WDBC, SEER	Improved diagnostic accuracy with ensemble; Weighted AUC mechanism for hybridization.	High computational cost and long training time.
Kemal P. et al.	MAD Normalization + KMC Feature Weighting + AdaBoostM1	Breast Cancer Coimbra (BCC)	Three-step hybrid approach achieved good accuracy.	Computationally expensive.
Teresa A. J. et al.	CNN on H&E-stained breast biopsy images	Bioimaging 2015 Challenge Dataset	Classified 4 medical categories with CNNs integrating multiple histological scales.	Not specified.
Fabio A. et al.	Patch-based CNN	BreaKHis	CNN trained on	Limited data leads

	Classification on Histopathological Images		image patches for efficient classification, avoids complexity.	to expensive experimentation.
Hiba A. et al.	Performance comparison: SVM, C4.5, Naive Bayes, k-NN	Wisconsin Breast Cancer datasets	Benchmarked 4 classifiers on WBC datasets.	No deep learning methods used; mainly classical ML algorithms.

III. Proposed Methodology

Breast cancer is one of the most common cancers among women worldwide. Early detection and diagnosis are crucial for effective treatment and improving survival rates. The Breast Cancer Wisconsin (Diagnostic) dataset contains features computed from digitized images of breast cancer biopsies. This dataset is used to train a machine learning model to predict the malignancy of tumors based on these features. This dataset is taken from Kaggle. The Breast Cancer Prediction Model is a machine learning project aimed at predicting whether a breast tumor is malignant or benign based on various diagnostic features. This project leverages data science and machine learning techniques to build a predictive model and deploy it as an interactive web application using Streamlit. This project has features like Dataset, Data Preprocessing, Exploratory Data Analysis (EDA), Model Building, Model Evaluation, and Deployment.

Steps to Build the Prediction Model:

1-Dataset

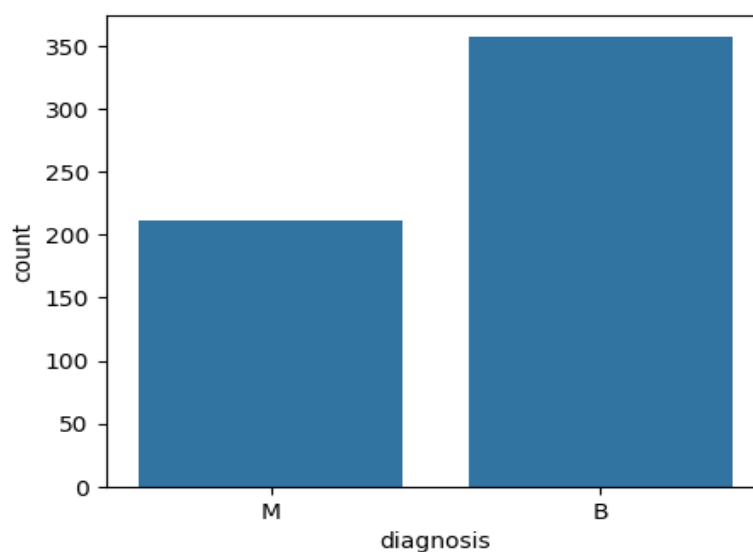
Breast cancer prediction involves developing a model to predict whether a breast cancer diagnosis is benign or malignant based on various features of the patient and tumor characteristics. Using the data.csv file, we can build a machine learning model for this purpose. The dataset is taken from Kaggle.

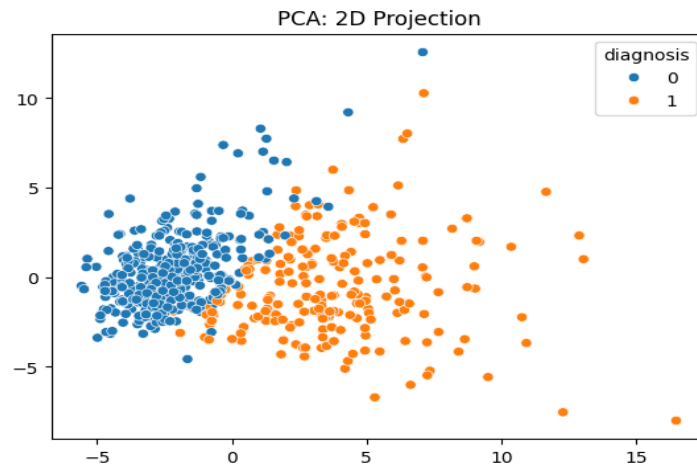
2-Data preprocessing

The data preprocessing steps were crucial for ensuring robust model training and included the following procedures:

- **Data Cleaning:** We addressed missing values and outliers and removed duplicate records to enhance the quality of the dataset. Categorical features were encoded using one-hot encoding to facilitate their use in the mathematical models.

Following is the **count plot**, and visualization of high-dimensional data with **PCA: 2D Projection** after cleaning the data.....





- **Feature Scaling:** All numerical features were scaled using the StandardScaler from scikit-learn to standardize the feature set, which is critical for the performance of many machine learning algorithms.
- **Data Splitting:** The dataset was split into a training set (80%) and a test set (20%) to evaluate the models effectively.

```

from sklearn.model_selection import train_test_split
[23] ✓ 12s Python

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
[24] ✓ 0.0s Python

X_train.shape
[25] ✓ 0.0s Python
... (455, 31)

X_test.shape
[26] ✓ 0.0s Python
... (114, 31)

```

Use `pandas` to clean, preprocess, and handle missing values.

Use `matplotlib` and `seaborn` for visualizing data distributions and relationships between features.

3-Model Selection

-Choose appropriate machine learning algorithms (e.g., Logistic Regression, Decision Tree, Random Forest, Support Vector Machine).

-Train multiple models and compare their performance.

4-Model Training and Evaluation:

-Train the selected models on the training data.

-Evaluate model performance using metrics.

Use `scikit-learn` for training and evaluating machine learning models, such as Logistic Regression, Random Forest, or Support Vector Machines.

5-Hyperparameter Tuning

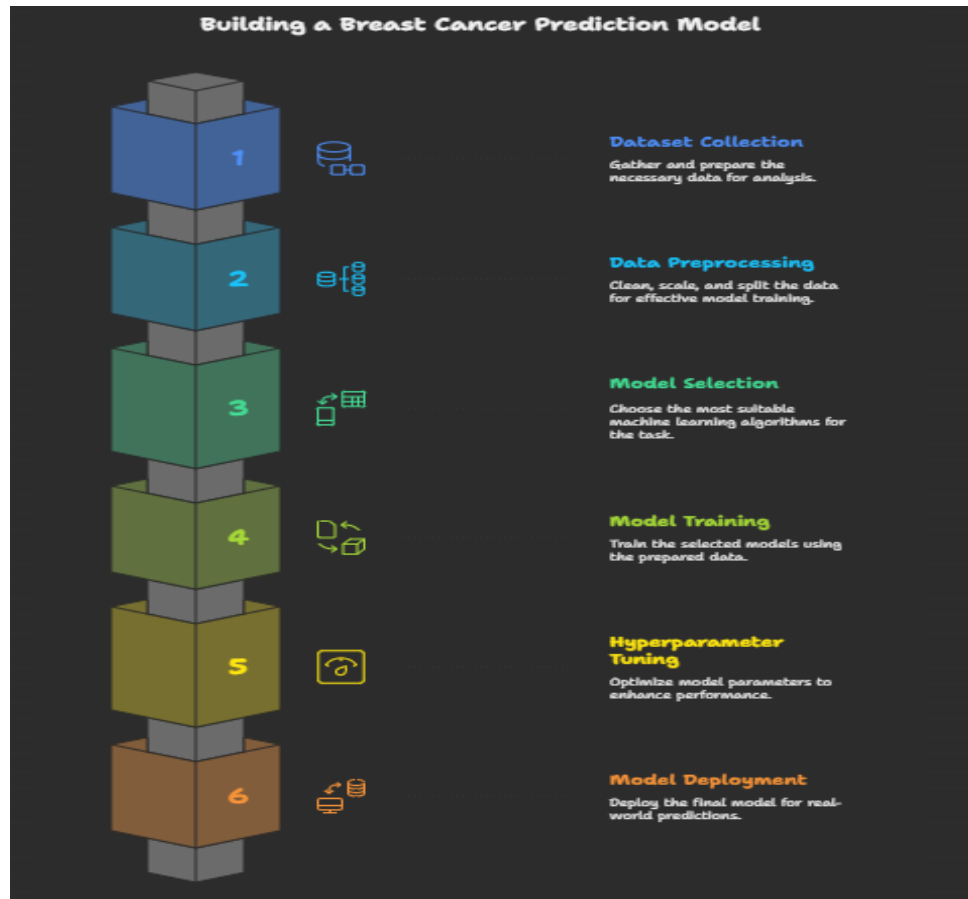
-Optimize model parameters using techniques like grid search or random search to improve performance.

6-Model Deployment

-Once the best model is selected, deploy it for predicting new data.

-Ensure the model is robust and can handle real-world data.

Deploy the model using Streamlit to create an interactive interface for users to input data and receive predictions.



Methadology

IV. Results

```
... logistic_regression Test Set Accuracy: 0.9912280701754386
    random_forest Test Set Accuracy: 0.9649122807017544
    svm Test Set Accuracy: 0.9736842105263158
    knn Test Set Accuracy: 0.9473684210526315
    decision_tree Test Set Accuracy: 0.9298245614035088

Logistics Regression Model is Best
```

The result shows the test set accuracy scores for five machine learning models used in a classification task are as follows:

1. **Logistic Regression:** Achieved the highest accuracy of *99.12%.
2. **Support Vector Machine (SVM):** Achieved an accuracy of *97.37%.

3. **Random Forest:** Recorded an accuracy of 96.49%.
4. **K-Nearest Neighbors (K-NN):** Attained an accuracy of 94.74%.
5. **Decision Tree:** Showed the lowest accuracy among the tested models at 92.98%.

In Contrast,

- Logistic Regression is identified as the model with the highest accuracy for this dataset.
- The results suggest that simpler models may generalize effectively on this dataset compared to more complex ones.
- SVM and Random Forest performed well, while K-NN and Decision Tree achieved lower accuracy scores.

This analysis highlights the importance of model selection, as different algorithms can produce varying levels of predictive performance.

V. Factors Impact on Prediction Models

Breast cancer is a complex disease with various factors affecting its development and progression. Machine learning (ML) techniques have been applied to predict breast cancer diagnosis and prognosis using patient data. This section discusses some of the critical factors that impact the accuracy and performance of breast cancer ML prediction models.

1. Quality and quantity of data

The quality and quantity of data have a significant influence on the model's accuracy. The dataset used should be representative of the population under study, with sufficient samples and features that capture the relevant clinical and molecular characteristics of breast cancer.

2. Feature selection and extraction

The selection and extraction of features are vital steps in developing accurate breast cancer prediction models. The selection of relevant features can improve the prediction accuracy while decreasing the dimensionality of the data. Feature extraction techniques can also be used to extract relevant features from high-dimensional data.

3. Choice of machine learning algorithm

ML algorithm is another critical factor in developing accurate prediction models, hence, selecting the appropriate model plays an important role in predicting breast cancer accurately. Popular ML algorithms have been used for breast cancer prediction. The choice of algorithm depends mainly on the dataset specifications, the complexity of the problem, and the performance metrics of interest.

4. Model evaluation and validation

Model evaluation and validation are crucial steps in developing accurate breast cancer prediction models. The performance of the model should be evaluated using appropriate metrics: sensitivity, specificity, accuracy, and area under the curve (AUC). The model should also be validated using independent datasets to assess its generalizability and robustness. Careful evaluation and validation of the model can ensure its accuracy and reliability for clinical applications.

5. Clinical relevance

The clinical relevance of the breast cancer prediction model is also an essential factor to consider. The model should be developed with the aim of improving clinical decision-making and patient outcomes. The model should also be interpretable by clinicians, providing insights into the biological and clinical factors that contribute to breast cancer growth and advancement.

In conclusion, the development of accurate breast cancer machine learning prediction models requires the consideration of multiple factors such as data quality and quantity, feature selection and extraction, machine learning algorithm, model evaluation and validation, and clinical relevance. The selection of appropriate strategies to address these factors can ameliorate the accuracy and clinical utility of the model.

VI. Discussion & Conclusion

In conclusion, applying a machine learning prediction model for breast cancer has shown promising results. Our research has provided an overview of the Kaggle dataset used to develop and test these models, highlighting their strengths and limitations. We found that machine learning models have the potential to improve breast cancer diagnosis and treatment, as they can accurately predict the likelihood of a patient having breast cancer, detect early signs of the disease, and help identify the most effective treatments for patients.

ML algorithms have demonstrated significant advantages in the prediction of breast cancer. Their ability to analyze vast amounts of data, detect patterns, and personalize treatment plans has significantly improved the accuracy of breast cancer prediction models.

However, there are still challenges that need to be addressed in the development and implementation of these models, including the need for large and diverse datasets, transparent and interpretable models, and ethical considerations related to data privacy and bias. Further research is needed to improve the accuracy and transparency of ML algorithms and to overcome the limitations in their application to breast cancer prediction. Addressing these challenges is crucial for realizing the full potential of ML in improving breast cancer diagnosis and treatment.

VII. Future Research Directions and Recommendations

While ML algorithms have demonstrated significant results in breast cancer prediction, there are several areas that require further research to improve their accuracy and reliability. One area that requires further investigation is the development of standardized protocols for data collection and analysis. Currently, there is a lack of consistency in the data used to train and test ML models. This can result in biased models that are not generalizable to diverse populations. Future research should aim to establish standard protocols for data collection, preprocessing, and feature extraction to ensure that the data used to train ML models is representative of the population.

Another area that requires further research is the incorporation of multiple modalities for breast cancer prediction. While ML algorithms have shown promising results using mammography images, genomic data, and patient demographics, the integration of these modalities can further improve the accuracy of breast cancer prediction models. Future research should explore the use of multimodal data fusion techniques to combine information from multiple sources and improve the accuracy of breast cancer prediction models.

The development of interpretable ML models is another area that requires further investigation. Currently, ML models are often considered “black boxes” as it is difficult to understand how they arrive at their predictions. This lack of interpretability can make it difficult for physicians to understand the rationale behind the model’s predictions and make informed decisions. Future research should aim to develop interpretable ML models that can provide physicians with explanations for their predictions, allowing for better clinical decision-making.

Finally, the development of ML models for predicting breast cancer recurrence and treatment response requires further investigation. While ML algorithms have shown promising results in predicting breast cancer diagnosis, their use in predicting treatment response and recurrence is limited. Future research should aim to develop ML models that can predict treatment response and recurrence, allowing physicians to tailor treatment plans and improve patient outcomes.

In conclusion, the development of ML models for breast cancer prediction requires further research to improve their accuracy, reliability, and interpretability. Future research should focus on developing standardized protocols for data collection and analysis, integrating multiple modalities, developing interpretable models, and predicting treatment response and recurrence. These research directions have the potential to improve the accuracy of breast cancer prediction models and improve patient outcomes.

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